

EXACT CHANNEL EXCHANGE AND JOINED DIVISOR WINDOWS FOR THE ERDŐS–STRAUS EQUATION

THE CLANKERS

ABSTRACT. We retain the original exterior and middle divisor channels. A reciprocal divisor involution exchanges them on its maximal integral domain. For $a = 2^e A$, A odd, the original packet with odd part dividing A has two explicitly overlapping cyclic windows, whose union has length $3e + 3$. If $a = \ell^e r$ has one simple nonresidue prime r and a residue prime ℓ , these become a complete channel classification, with an exact one-unit count comparison when 4 belongs to the power subgroup. The theorem is an exact arithmetic comparison on its specified factor class, not universal ES occupancy.

Let $p \equiv 1 \pmod{4}$ be prime, $p/4 < a < p$, $R = 4a - p$, and $\gcd(pa, R) = 1$. The original labelled gates for $u \mid a^2$ are

$$E : R \mid 4u + 1, \quad M : R \mid u + a.$$

Put $d = \gcd(a, u)$ and $(h, r, s) = (d^2/u, u/d, a/d)$. The ordered returns are

$$E : (a, h s \kappa, p h r \kappa), \quad \kappa = (p r + s)/R; \quad M : (a, p h s \lambda, p h r \lambda), \quad \lambda = (r + s)/R.$$

Prime valuations give $a = h r s, u = h r^2, \gcd(r, s) = 1$; multiplication verifies $4/p$. The ordered inverses are $a^2/(Ry - pa)$ and $pa^2/(Ry - pa)$. This arithmetic is inherited Type I/II parametrization [1].

Proposition 1 (Exact channel involution). *If $4 \mid a$, the map $u \mapsto a/(4u)$ is an involution of $\text{Div}(a/4)$ and interchanges E and M . This is its maximal integral domain among original divisors of a^2 .*

Proof. Integrality is exactly $4u \mid a$. Put $v = a/(4u)$. Since a, u are units modulo R , the equality $a(4u + 1) = 4u(v + a)$ proves the equivalence of the gates. The ratio $v/u = a/(4u^2)$ has quadratic character (a/p) , so it actually changes character on a negative source. A numerical fixed point requires $a = 4u^2$; its E and M labels remain distinct. For example, $p = 61, a = 16, R = 3, u = 2$ is fixed. $\square$

Write $a = 2^e A$, A odd, $e \geq 1$. Retain the actual subpacket $u = 2^f d$, $0 \leq f \leq 2e$, $d \mid A$. Its omitted odd-square divisors are not removed from the full source. For $w \mid A$ consider

$$I_E = [-2e - 2, -2], \quad I_M = [-e, e], \quad 2^j = -w \pmod{R}.$$

Theorem 2 (Joined windows). *The solutions in these integer intervals are in bijection with the original packet hits, by*

$$E : u = 2^{-j-2} w, \quad M : u = 2^{e+j} A/w.$$

The inverse reads $f = v_2(u)$ and the odd part d and gives $(w, j) = (d, -f - 2)$ or $(A/d, f - e)$ respectively. Their union is $[-2e - 2, e]$, of length $3e + 3$. Their overlap is $[-e, -2]$ and pairs precisely the involution above.

Let $D = \text{ord}_R(2)$ and $A_- = \#\{w \mid A : -w \in \langle 2 \rangle_R\}$. If $3e + 3 \geq D$, the packet is jointly occupied exactly when $A_- > 0$. Write $2e + 1 = f_0 D + \rho$, $0 \leq \rho < D$, and set $\delta = \min((e + 2) \bmod D, (-e - 2) \bmod D)$. Its tagged joint count satisfies

$$A_-(2f_0 + \mathbf{1}_{\rho \geq D-\delta}) \leq C \leq A_-(2f_0 + \mathbf{1}_{\rho > 0} + \mathbf{1}_{\rho > \delta}).$$

The two constants are the exact extrema over cyclic target classes.

Proof. E is equivalent to $2^{-f-2} = -d$, while M is equivalent to $2^{f-e} = -A/d$. These equalities prove the bijections. The intervals overlap or are adjacent when $e \geq 1$, proving the union. The product of their overlap words is $a/4$. For counts, remove f_0 complete cycles from each interval. The remaining arcs have length ρ and separation with shorter distance δ . They overlap exactly when $\rho > \delta$, and cover the cycle exactly when $\rho \geq D - \delta$. Their indicator-sum extrema give the bounds. Finally a consecutive interval of length $3e + 3$ fills the cycle exactly when that length is at least D . $\square$

For $e = 0$ the intervals are the two singletons $\{-2\}, \{0\}$; the consecutive union assertion is not used. Logarithms specify residue classes; the returned integer exponents must still lie in the original intervals.

Theorem 3 (Complete counts with one simple nonresidue factor). *Suppose $a = Br$, $r \nmid B$ is prime, $(r/p) = -1$, and every prime factor of B has character $+1$. Put $\mu_B(g) = \#\{v \mid B^2 : vB^{-1} = g \pmod{R}\}$. Then*

$$\boxed{|E_a| = \mu_B(-p^{-1}), \quad |M_a|/2 = \mu_B(-r).}$$

For $B = \ell^e$ with $4 \in \langle \ell \rangle_R$, let its order be D and retain $c = \log_\ell 4$. If $-r$ lies in the subgroup with $\log j$, then

$$|E_a| = \#\{n \in [-2e - c, -c] : n = j \pmod{D}\}, \quad |M_a|/2 = \#\{n \in [-e, e] : n = j \pmod{D}\}.$$

In particular $||E_a| - |M_a|/2| \leq 1$. If $-r$ is outside, both vanish. If 4 is outside the power subgroup, at most one channel is occupied.

Proof. For every original square divisor, quadratic reciprocity gives $(u/R) = (u/p)$: for each odd prime $t \mid a$, $R = -p \pmod{t}$ and $R = 3 \pmod{4}$ cancel the two signs; the supplementary law handles 2. E therefore requires $(u/p) = -1$, and M requires $(u/p) = -(a/p) = 1$. The permitted exponent of r is consequently one in E and zero or two in M. The E word is $u = rv$; the first M orientation is $u = v$, and its complement a^2/v is the other. This proves the coefficient formulas with all multiplicities. For $B = \ell^e$, the E gate becomes $\ell^{-f-c} = -r$, and the first M gate becomes $\ell^{f-e} = -r$. The two intervals have the same length, hence their counts of one residue differ by at most one. When 4 is outside, the necessary cosets for $-r$ are $4^{-1}\langle \ell \rangle$ and $\langle \ell \rangle$, which are disjoint. $\square$

For $\ell = 2$, one may take the integer lift $c = 2$. Thus the joined-window theorem classifies the *whole* count $|E| + |M|/2$ in this factor class. At the hard prime 23209, $R = 127$, $a = 2 \cdot 2917$, the order is seven and the log of -2917 is two. The E classes are 3, 4, 5 and the M classes 6, 0, 1; both complete channels miss. This realizes the one-class shortfall from the universal cyclic-cover threshold. At residual 31 the primes 43801, 8521, 12601 realize respectively $(|E|, |M|/2) = (1, 0), (0, 1), (1, 1)$, so the count difference cannot be improved to equality.

The complete six-vertex stopping example, its original source factorizations and ordered returns are recorded in the accompanying workbench and executable certificates. The window inequalities do not assert that every prescribed prime supplies a suitable factorization or an occupied original target.

REFERENCES

- [1] C. Elsholtz and T. Tao (2013), *Counting the number of solutions to the Erdős–Straus equation on unit fractions*, J. Austral. Math. Soc. 94, 50–105; arXiv:1107.1010, §2, Propositions 2.2 and 2.6. Used parametrizations are rederived.
- [2] The Clankers (2026), *Turn 4: mixed square-source finite reduction*, PR19 revision 4b6bb0f187fa44bb62fc8c51e996f05dfc90e2d7. The source and factor-expansion results do not supply selector occupancy.